Self-Check Questions: Introductory Analysis Supplement

Prerequisite review: real sequences

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Scope. These supplementary questions review background assumed by John K. Hunter's *An Introduction to Real Analysis* (2012), especially Chapter 1. They were not extracted from that lecture chapter and are separate from the source-based questions in `intro-analysis-self-check.tex`.

How to use these questions. Recall definitions and basic properties, including their hypotheses and quantifiers. Full proofs are not required for this first pass. No solutions are included.

All sequence terms and finite limits are real numbers unless specified otherwise. Use $\mathbb{N} = \{1, 2, 3, \ldots\}$.

1 Sequences and convergence

1. Define a real sequence as a function. How does a sequence differ from the set of its values? May a sequence repeat a value?
2. State the definition of $x_n \to L$ using ϵ and N, with all quantifiers in order. What does it mean for a condition to hold eventually?
3. In the definition of convergence, may N depend on ϵ? May it depend on the index n quantified after it? Must one value of N work for every $\epsilon > 0$?
4. Write the precise negation of $x_n \to L$ for a specified $L \in \mathbb{R}$. How does this differ from saying that the sequence has no finite limit?
5. Can a real sequence have two different finite limits?
6. Does changing or deleting finitely many initial terms affect convergence or the value of a limit?
7. Define a bounded sequence using an absolute-value inequality. Is it enough for all sufficiently late terms to satisfy such an inequality?
8. Is every convergent real sequence bounded? Is every bounded real sequence convergent? Recall a simple counterexample to any implication that fails.

2 Subsequences

9. Define a subsequence (x_{n_k}) of (x_n). What conditions must the indices n_k satisfy? Must $n_k \to \infty$? May equal values occur in a subsequence?
10. If $x_n \to L$, what can be said about every subsequence? Must all terms of a subsequence be different from L?

11. Does the existence of one convergent subsequence imply that the original sequence converges? Recall a simple counterexample if it does not.

12. What can be concluded if a sequence has two subsequences converging to different finite limits?

3 Basic limit properties

13. Suppose $x_n \to x$, $y_n \to y$, and $c \in \mathbb{R}$. State the limit rules for $x_n + y_n$, cx_n, $x_n y_n$, and x_n/y_n. What extra hypotheses are needed for the quotient?

14. Suppose $x_n \to x$, $y_n \to y$, and eventually $x_n \le y_n$. What inequality follows for the limits? If eventually $x_n < y_n$, must the limit inequality be strict?

15. State the squeeze (or sandwich) theorem for real sequences, including the required inequalities and limits.

16. Define nondecreasing and nonincreasing sequences. How do these differ from strictly increasing and strictly decreasing sequences?

17. State the monotone convergence theorem for real sequences. Which one-sided bound is needed in each case? Express the limit as a supremum or infimum of the sequence's values.

18. Define $x_n \to +\infty$ and $x_n \to -\infty$ using quantifiers. Does every sequence with no finite limit diverge to one of these infinities? Recall a simple counterexample if it does not.